

DIHWorld Audio

Dynamics / Guard Technical Manual

Current product versions, controls, presets, meters, and placement

DIHWorld Audio builds connected production systems around the Imhotep 369 Engine—a Triadic Harmonic Lattice—and Ma'at 357 / MomentumBend. Phi-derived control and the canonical Phi/2 ceiling preserve movement, balance, and musical intent without brickwall limiting.

This edition documents 4 current products. Release numbers are taken from the installed DIHWorld Audio Audio Units verified on 2026-09-05; they are product versions, not a shared catalog version.

Canon reference: normalized $\Phi/2 = 0.809016994$. MomentumBend shapes motion and containment; the ceiling is a canonical boundary, not permission for hard limiting.

Operating law: begin from reset or a purpose-matched preset, compare at equal loudness, change one system at a time, and judge the result in the full arrangement.

ANUBIS

RELEASE	FAMILY	PLACEMENT
v1.0.3	Dynamics / Guard	Insert Safe

Dynamics gate and guard with threshold, ratio, attack, hold, release, range, a named stereo host Sidechain bus, independent Sidechain Key and Sidechain Duck operation, protected I/O, and parallel mix.

Recommended use: guard/dynamics

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
input	Input	Float
mix	Mix	Float
output	Output	Float
power	Power	Bool
threshold	Threshold	Float
ratio	Ratio	Float
attack	Attack	Float
hold	Hold	Float
release	Release	Float
range	Range	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
extkey	Sidechain Key	Bool
sduck	Sidechain Duck	Bool
gainmatch	Gain Match	Bool

Factory starting points

TRANSPARENT PASS [Stereo] • VOCAL BREATH GUARD [Stereo] • KICK KEY GATE [Stereo] • TEHUTI DUCK GUARD [Stereo] • KEY + DUCK LOCK [Stereo] • DRUM TIGHT GUARD [Stereo] • MONO WALL HOLD [Mono] • DUAL MONO LOCK [Dual Mono]

Meters and visualizers

IN: Main-input RMS energy. Shows the program feeding the gate.

GATE: Current gate passage gain. OPEN is full passage; lower values mean attenuation.

KEY: External sidechain RMS energy. Moves only when a host sidechain is routed.

DUCK: Sidechain-driven gain reduction. More fill means deeper ducking of the main signal.

OUT: Post-gate RMS energy. Compare with IN to judge actual attenuation.

Transfer graph: Threshold, envelope, and Range floor. Shows where the signal opens and how far closure may fall.

Correlation waves: Left/right phase relationship. Gold is coherent, purple is spread, red warns of anti-phase risk.

OTIS

RELEASE	FAMILY	PLACEMENT
v2.1.16	Dynamics / Guard	Insert Safe

3-6-9 macro dynamics processor: Micro Peak controls short events, Meso Density shapes body, and Macro Level governs the long envelope.

Recommended use: dynamics support; secondary silver mastering-support dot

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
trumpet_l	Micro Peak	Float
trumpet_c	Meso Density	Float
trumpet_r	Macro Level	Float
makeup_gain	Makeup Gain	Float
gainmatch	Gain Match	Bool

Factory starting points

TRANSPARENT PASS [Stereo] • VOCAL LEVELER [Stereo] • DIALOG STEADY [Stereo] • DRUM BUS CATCH [Stereo] • MIX BUS GLUE [Stereo] • TRANSIENT GUARD [Stereo] • MONO LEVEL HOLD [Mono] • DUAL MONO CONTROL [Dual]

Meters and visualizers

INPUT: Input peak in dB. Confirms level feeding all three time scales.

RESPONSE: Strongest active 3/6/9 reduction. Shows the greatest current reduction across Micro, Meso, or Macro.

OUTPUT: Output peak in dB. Use with INPUT and Gain Match for fair comparison.

3 / Micro activity: Fast transient reduction. Responds to short peaks and front edges.

6 / Meso activity: Body and phrase-density reduction. Responds to medium-duration energy.

9 / Macro activity: Long-envelope reduction. Responds to sustained level movement.

DeBo

RELEASE	FAMILY	PLACEMENT
v3.6.9	Dynamics / Guard	Insert Safe

Pressure-activated dynamics processor with fourteen canonical programs: pressure progression plus purpose-built Vocal, Drum, Mix Bus, Parallel, and Master programs. The approved powered-on default is unchanged; the named bank uses a warmer depth/pressure progression without runtime preset clamping, topping out at 2.2 dB Maximize.

Recommended use: pressure-activated dynamics guard

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
POWER	Power	Bool
FLOWER	Flower of Life	Bool
GAINMATCH	Gain Match	Bool
INPUT	Input	Factory
DRIVE	Drive	Factory
MAXIMIZE	Maximize	Factory
DEPTH	Depth	Factory
CLARITY	Clarity	Factory
OUTPUT	Output	Factory

Factory starting points

NO BRICKS [Stereo] • VOCAL HOLD [Stereo] • BRICK TAP [Stereo] • WALL PRESSURE [Stereo] • SHOULDER CHECK [Stereo] • NO SCISSORS [Stereo] • BREAKTHROUGH [Stereo] • RED LINE WATCH [Stereo] • PHI GUARD [Stereo] • VOCAL ANCHOR [Stereo] • DRUM REINFORCE [Stereo] • MIX BUS SETTLE [Stereo] • PARALLEL BREACH [Stereo] • MASTER FIELD [Stereo]

Meters and visualizers

RMS: Sustained signal body. Shows average energy driving the pressure model.

GUARD PRESSURE: Combined density/containment demand. Higher values mean DEBO is working harder.

PEAK: Fast peak relative to ceiling. Approaching the top warns of limited remaining space.

GOVERNOR: Gain held back by protection. More activity means stronger safety containment.

Pressure window / face: RMS-driven pressure stage. The visual advances with real pressure; red is the ceiling warning state.

LookOut

RELEASE	FAMILY	PLACEMENT
v1.0.1	Dynamics / Guard	Insert Safe

Dual-envelope transient/sustain shaper. Attack and Sustain directly move the level of the transient and sustain portions of the signal, up to +/-18 dB each, ahead of the DIH tone-color stage. Its Eye of Heru state also gives live advisory guidance on transient/sustain balance while it shapes.

Recommended use: dual-envelope transient/sustain shaper; Attack/Sustain directly move transient and sustain gain up to +/-18dB; Eye of Heru advisory guidance is secondary, not the primary function; cyan primary dot with green observation and silver mastering-support dots

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
power	Power	Factory
attack	Attack	Factory
sustain	Sustain	Factory
gainmatch	Gain Match	Factory

Factory starting points

NEUTRAL SENTINEL [Stereo] • VOCAL FRONT GUARD • DRUM SNAP LOCK • KICK BODY ANCHOR • PERCUSSION EDGE • ROOM TAIL WATCH • MIX BUS COMPOSURE • FINAL PEAK SENTINEL

Meters and visualizers

Transient: Detected front-edge energy. Use to judge whether Attack is emphasizing or restraining hits.

Sustain: Detected body and tail energy. Use to judge Sustain changes to ring, room, and body.

Alert: Potential transient/sustain imbalance. Higher values suggest the balance needs attention.

Miss: Underrepresented event energy. Indicates material that may be disappearing.

Stress: Over-concentrated envelope pressure. Reduce shaping intensity when stress remains high.

Composure: Stable transient/sustain balance. Higher is calmer and more controlled.

Eye of Heru: Combined advisory state. It reports balance; it does not move controls automatically.